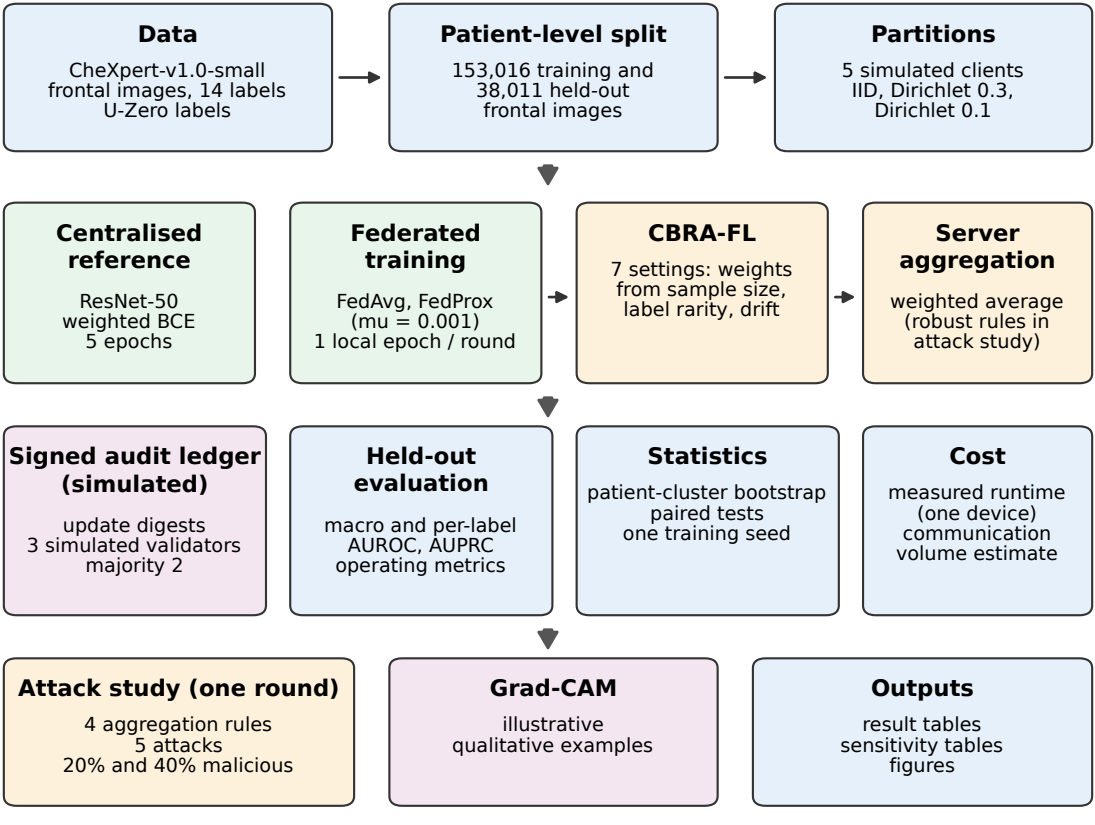
Data
CheXpert-v1.0-small
frontal images, 14 labels
U-Zero labels
Patient-level split
153,016 training and
38,011 held-out
frontal images
Partitions
5 simulated clients
IID, Dirichlet 0.3,
Dirichlet 0.1
Centralised reference
ResNet-50
weighted BCE
5 epochs
Federated training
FedAvg, FedProx
(mu = 0.001)
1 local epoch / round
CBRA-FL
7 settings: weights
from sample size,
label rarity, drift
Server aggregation
weighted average
(robust rules in
attack study)
Signed audit ledger (simulated)
update digests
3 simulated validators
majority 2
Held-out evaluation
macro and per-label
AUROC, AUPRC
operating metrics
Statistics
patient-cluster bootstrap
paired tests
one training seed
Cost
measured runtime
(one device)
communication
volume estimate
Attack study (one round)
4 aggregation rules
5 attacks
20% and 40% malicious
Grad-CAM
illustrative
qualitative examples
Outputs
result tables
sensitivity tables
figures